

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer at BMO's AI Centre of Excellence. Built BMO Wealth's multi-agent data layer from the ground up, scaled it, and own it today: an orchestration harness giving agents graph, analysis, and search tools over client data. Built the eval pipeline that caught a production GenAI tool serving \$200B+ AUM downplaying investment risk. Took a proprietary subcutaneous neurostimulator from idea to production device at Epineuron. Co-founder of Merlyn Labs; open-sourced a flow-matching VLA integration for RLinf.

CORE COMPETENCIES

Agentic Systems

Agent Evaluation

LLM Fine-tuning

Reinforcement Learning

Python & C/C++

PyTorch & JAX

WORK EXPERIENCE

BMO AI Centre of Excellence Sep 2025 - Present

AI Research Engineer

Toronto, Canada

- Built, scaled, and own BMO Wealth's multi-agent data layer: an orchestration harness with graph, analysis, and search tools.
- Example capability: agents answer multi-hop relational queries over a graph database of the bank's client data.
- Built a deterministic eval pipeline that caught a production GenAI tool serving \$200B+ AUM downplaying investment risk.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder

Toronto, Canada

- Open-sourced a flow-matching VLA integration for RLinf, enabling RL training on the BEHAVIOR-1K suite in OmniGibson.
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game.
- Showed $\pi 0.5$'s LIBERO-PRO failures are recipe-induced overfitting; proposed a recalibrated baseline for the benchmark.
- Conservative finetuning doubled position-swap success (21% to 42%) with no architectural change to the model.

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga, Canada

- Owned a proprietary subcutaneous neurostimulator from first idea to production device at a clinical-stage startup.
- Designed and assembled the PCBs and wrote the IEC 60601-1 / ISO 13485 validation protocols for clinical test runs.
- Part of the team behind PeriPulse, an FDA Breakthrough-designated device now in multinational clinical trials.
- Extended battery life 900% through oscilloscope and power-analyzer testing of the device's power draw.
- Modeled electromagnetic field penetration into peripheral nerves in COMSOL; the results set the electrode diameter.

PROJECTS

Stanford BEHAVIOR-1K Challenge 8th Place, Standard Track

Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling 3x. Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report and a LessWrong analysis.

BardSong Closed Alpha

Built the end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video. Fine-tuned an LLM on D&D sourcebooks for narrative coherence; in closed alpha with 23 dungeon masters.

EDUCATION

M.Eng in AI & Robotics University of Toronto Sep 2024 - Apr 2027 (anticipated)

B.Eng, Engineering Physics Co-op McMaster University Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Agents & ML: Agents, Agent Evaluation, Alignment Evaluation, LLM Fine-tuning, Reinforcement Learning, Imitation Learning, Flow Matching
Systems & Languages: Python, C/C++, PyTorch, JAX, OmniGibson, MuJoCo, ROS2, Sim-to-Real, PCB Design